

Deep Multi-Omics Fusion for Breast Cancer Molecular Subtyping and Recurrence Risk Prediction: A Schema-Matched Synthetic Proof-of-Concept

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Highlights

- A deep multi-branch fusion architecture integrated genomic, transcriptomic, proteomic and methylation data.
- Multi-omics fusion matched PAM50 expression-based classification (macro-AUC 0.992 vs 0.999).
- Every individual omics layer, including genomics alone (AUC 0.823), classified subtype above chance.
- Bootstrap analysis identified a stable 16-feature multi-omics signature spanning all four layers.
- Multi-omics fusion did not significantly outperform a PAM50-subtype-plus-clinical baseline for 5-year recurrence in this synthetic cohort ($\Delta\text{AUC} = -0.003$, $p = 0.42$).

Abstract

Breast cancer molecular subtyping and recurrence-risk assessment rely heavily on PAM50 gene-expression profiling, but real tumors carry genomic, epigenetic and proteomic alterations that a single expression panel does not capture. We developed a deep multi-branch fusion framework that integrates four omics layers -- somatic mutation and copy-number status (genomics), a 90-gene expression panel including the full PAM50 set (transcriptomics), a 45-antibody reverse-phase protein array panel (proteomics), and promoter CpG methylation at 35 loci (methylation) -- for (i) four-class molecular subtype classification and (ii) five-year recurrence risk prediction. Each omics layer is compressed by a small per-branch neural sub-network, and branch representations are combined by a fusion head. The framework was mechanically validated on a synthetic, schema-matched cohort ($n = 520$) constructed so that every feature across all four layers derives from shared, patient-level latent biology (estrogen-receptor, HER2, proliferation and basal activity, plus a subtype-only-partially-coupled genomic instability axis), giving a checkable ground truth. Five-fold cross-validation (8 repeats) showed near-ceiling four-class subtype discrimination from PAM50 expression alone (macro-AUC = 0.999) and from the deep multi-omics fusion model (macro-AUC = 0.992); every individual omics layer, including genomics alone, classified subtype above chance (AUC range 0.823-0.999). Bootstrap stability analysis (80 resamples, 70% threshold)

identified a reproducible 16-feature multi-omics signature spanning all four layers, whose members correlated with their expected estrogen-receptor, HER2, proliferation or basal-pathway markers in the predicted direction. For five-year recurrence prediction, deep multi-omics fusion combined with clinical covariates achieved AUC = 0.801 versus 0.804 for a PAM50-subtype-plus-clinical baseline, a difference not statistically significant in this cohort (paired Δ AUC = -0.003, Wilcoxon p = 0.42). This proof-of-concept demonstrates that the fusion architecture integrates heterogeneous, high-dimensional multi-omics data without loss of subtyping performance and recovers a biologically coherent signature, but did not demonstrate a recurrence-prediction advantage over PAM50 subtype in this synthetic design; we discuss why, and what a real-data application using TCGA-BRCA and METABRIC would require to test the "beyond PAM50" hypothesis properly.

Keywords

Multi-omics integration; Breast cancer; Deep learning; Molecular subtyping; Recurrence prediction; Data fusion

1. Introduction

Breast cancer is molecularly heterogeneous, and its clinical management is shaped by intrinsic subtype -- luminal A, luminal B, HER2-enriched, and basal-like -- first defined by gene-expression clustering [1,2] and later operationalized as the clinically used PAM50 classifier [3]. The Cancer Genome Atlas (TCGA) subsequently profiled hundreds of breast tumors across genomic, transcriptomic, proteomic and epigenetic layers simultaneously, showing that each molecular layer carries partially distinct, partially overlapping information about a tumor's biology [4], and the METABRIC study extended this to a cohort of roughly 2,000 breast tumors with integrated copy-number and expression data linked to long-term clinical outcome [5]. Despite this, PAM50 subtyping and most clinical recurrence-risk tools still rely predominantly on a single expression panel, leaving genomic (mutation, copy-number), epigenetic (methylation) and proteomic alterations largely unused at the point of clinical decision-making.

Multi-omics integration -- combining two or more molecular data layers from the same patients into a single analysis -- aims to recover this unused information. Classical statistical approaches include joint latent-variable models such as iCluster [6] and network-based approaches such as similarity network fusion, which builds a patient-similarity network per omics layer and fuses them into a single network for downstream clustering [7]. More recent unsupervised approaches, such as Multi-Omics Factor Analysis, learn a small number of latent factors that jointly explain variation across omics layers, analogous to a multi-view extension of factor analysis or PCA [8]. A systematic benchmark of multi-omic clustering algorithms across TCGA cancer types found that no single integration strategy dominates uniformly, and that the choice of method materially affects the cancer subtypes recovered [9].

Deep learning offers an alternative integration strategy: rather than hand-designing a joint statistical model, a neural network can be trained to compress each omics layer into a shared or per-layer latent representation, optimized end-to-end for a downstream task. This approach has shown promise for survival stratification -- for example, autoencoder-based fusion of methylation, mRNA and miRNA data stratified hepatocellular carcinoma patients into prognostically distinct groups more effectively than single-omics or clustering-based baselines [10] -- and for treatment-response prediction, where late-integration neural architectures that encode each omics layer separately before combining them have outperformed early (feature-concatenation) integration and single-omics baselines for drug-response prediction using cell-line and patient-derived data [11]. These results motivate testing whether an analogous deep multi-branch fusion architecture can improve breast cancer molecular subtyping and, more ambitiously, recurrence-risk prediction beyond what PAM50 subtype alone provides -- since a large share of clinically relevant heterogeneity in local and distant recurrence risk is known to occur within, not just between, PAM50 subtypes, which is precisely the rationale for supplementary genomic recurrence-risk assays used alongside intrinsic subtyping in current practice.

This report has two aims. First, we implement and mechanically validate a deep multi-branch fusion architecture -- per-omics neural sub-networks feeding a fusion head -- for four-class molecular subtype classification and for five-year recurrence prediction, using a synthetic, schema-matched cohort constructed so that genomics, transcriptomics, proteomics and methylation features share checkable, literature-consistent biological structure with a known ground truth. Second, and more specifically, we test whether this fusion architecture predicts recurrence beyond a PAM50-subtype-plus-clinical baseline within that synthetic design, and use the result -- whichever direction it points -- to clarify what a real-data application to TCGA-BRCA and METABRIC would need to demonstrate a genuine "beyond PAM50" advantage.

2. Materials and methods

2.1. Study workflow

The framework has five stages, implemented in Python 3.12 using numpy [23], pandas, scipy [25], and scikit-learn [24], with figures produced in matplotlib [26]: (i) simulation of a schema-matched, paired multi-omics cohort (genomics, transcriptomics, proteomics, methylation, clinical) with known underlying biology; (ii) five-fold cross-validated four-class molecular subtype classification, benchmarked across PAM50-expression-only, each single omics layer alone, naive feature concatenation, and a deep multi-branch fusion architecture; (iii) five-fold cross-validated five-year recurrence prediction, benchmarked between a PAM50-subtype-plus-clinical baseline and a deep multi-omics fusion-plus-clinical model, including a paired significance test across folds; (iv) bootstrap permutation-importance stability analysis across all 187 candidate multi-omics features to derive a robust multi-omics signature; and (v) correlation analysis between signature features and canonical ER, HER2, proliferation and basal marker genes/proteins for biological validation. Full source code is provided with the manuscript (Code availability).

2.2. Synthetic schema-matched multi-omics cohort

Because this report validates the framework's mechanics rather than reporting findings from real patient data, a synthetic cohort ($n = 520$) was constructed with a schema deliberately matched to public multi-omics resources such as TCGA-BRCA and METABRIC [4,5]: (a) a genomics table of 15 driver-gene mutation/amplification indicators (TP53, PIK3CA, GATA3, MAP3K1, CDH1, PTEN, AKT1, RB1, MYC amplification, ERBB2 amplification, BRCA1 loss-of-function, BRCA2 loss-of-function, NF1, ARID1A, CCND1 amplification) plus tumor mutational burden; (b) a transcriptomics table of 90 genes, comprising the full 50-gene PAM50 panel [3] plus 40 additional expression features; (c) a proteomics table of 45 reverse-phase protein array-style analytes (total and phospho-proteins spanning ER/PR/HER2, PI3K-AKT-mTOR, MAPK, cell-cycle, apoptosis and epithelial-mesenchymal markers); and (d) a methylation table of 35 promoter CpG probes, including BRCA1, ESR1, CDH1, APC and other tumor-suppressor promoters with established silencing relevance in breast cancer. All four omics tables and a clinical table (age, tumor size, nodal status, five-year recurrence) were generated for the same 520 simulated patients, sharing four patient-level latent biological axes -- ER/luminal activity, HER2 activity, proliferation, and basal activity -- plus a separate genomic instability axis. Four molecular subtypes (luminal A, luminal B, HER2-enriched, basal-like) were assigned per patient (prevalence 39.8%, 20.0%, 16.9%, 23.3%, respectively, approximating TCGA-BRCA/METABRIC proportions [4,5]) and every downstream feature -- every gene, protein, CpG probe and mutation indicator -- was generated as a noisy function of these shared latent axes, so that every feature carries real, checkable common biological structure while retaining substantial feature-specific and patient-specific noise. Critically, the genomic instability axis was deliberately only weakly coupled to subtype (narrow between-subtype separation, wide within-subtype spread), so that a meaningful share of genomic and epigenetic variation -- and, in turn, of five-year recurrence risk -- is subtype-independent by construction; this is what makes it possible, in principle, for a multi-omics model to add recurrence-relevant information beyond a PAM50-subtype label. Five-year recurrence status was simulated from a logistic model combining subtype, continuous proliferation activity, clinical covariates (age, tumor size, nodal status), and a residual-risk term built from BRCA1/BRCA2 loss-of-function status, TP53 mutation, BRCA1 promoter hypermethylation, tumor mutational burden and genomic instability -- an explicit, checkable stand-in for the kind of subtype-independent genomic risk information that real

recurrence-risk assays are designed to capture. Full generative code, including the exact feature-to-latent-axis mapping and random seed, is in `data_simulation.py`.

2.3. Deep multi-branch fusion architecture for molecular subtyping

Seven approaches were benchmarked for four-class subtype classification using five-fold stratified cross-validation, repeated across 8 random seeds (40 folds total): (a) PAM50 expression alone (50-gene panel, multinomial logistic regression), representing the current expression-based clinical standard; (b)-(e) each single omics layer alone (genomics, transcriptomics, proteomics, methylation), using a random forest classifier (300 trees, maximum depth 6) [15]; (f) naive concatenation of all four layers into one feature matrix (187 features), classified by gradient boosting (200 stages, maximum depth 3) [16]; and (g) a deep multi-branch fusion architecture, in which each omics layer is independently standardized and passed through its own one-hidden-layer neural encoder (16 hidden units, rectified-linear activation, L2-regularized, trained with early stopping) [17,18], the four branches' hidden-layer activations are concatenated into a fused representation, and a two-hidden-layer fusion head (64-32 units) is trained on the fused representation to predict subtype. Out-of-fold predicted class-probabilities were used to compute macro-averaged one-vs-rest AUC for every approach.

2.4. Deep multi-branch fusion architecture for recurrence prediction beyond PAM50

Two approaches were benchmarked for five-year recurrence prediction, again using five-fold stratified cross-validation repeated across 8 seeds (40 paired folds): (a) a PAM50-subtype-plus-clinical baseline, representing current standard-of-care practice (categorical intrinsic subtype plus age, tumor size and nodal status, logistic regression); and (b) a deep multi-omics fusion-plus-clinical model, in which each omics layer is passed through a compact per-branch risk sub-network -- a one-hidden-layer neural network (6 hidden units, strongly L2-regularized given the modest per-fold sample size) supervised directly on the recurrence label -- and each branch's hidden-layer embedding together with its own scalar predicted-risk score is combined with clinical covariates by a single L2-regularized logistic regression fusion head. Supervising the branch sub-networks directly on recurrence (rather than on subtype) was a deliberate design choice: it allows subtype-independent residual-risk information -- e.g., BRCA1/2 loss-of-function status or BRCA1 promoter hypermethylation -- to be retained in the branch representations rather than discarded, which is the specific mechanism by which a multi-omics model could, in principle, add information beyond a PAM50-subtype label. The fusion head's capacity (a single logistic regression layer, rather than a further deep network) was kept deliberately low relative to fold sample size (~416 training patients per fold) to limit overfitting. Performance was compared both by independent-sample summary statistics (mean AUC $\pm$ SD across the 40 folds per arm) and by a paired analysis across the 40 matched folds (identical train/test splits for both arms within a given repeat), using a paired t-test and a Wilcoxon signed-rank test on the per-fold AUC differences.

2.5. Bootstrap multi-omics signature stability analysis

Following the same bootstrap-stability logic used in explainable-AI biomarker discovery generally [19], each of 80 bootstrap resamples of the cohort (stratified by subtype) was used to refit a random forest (150 trees, maximum depth 7) on all 187 candidate multi-omics features (16 genomics, 90 transcriptomics, 45 proteomics, 35 methylation, excluding one non-informative genomics count column) and record each feature's Gini importance. Each feature's stability frequency was defined as the fraction of the 80

resamples in which it ranked among the top 20 most important features; features with a stability frequency of at least 0.70 were retained as the robust multi-omics signature.

2.6. Signature-marker biological validation

For every signature feature, Pearson correlation across all 520 patients was computed against four canonical marker features representing the ER/luminal (ESR1), HER2 (ERBB2), proliferation (AURKA) and basal (KRT5) axes (16 features x 4 markers = 64 tests), with Benjamini-Hochberg false discovery rate correction applied across all 64 p-values [22], testing whether the framework's data-driven signature recovers the specific, literature-motivated directional associations it was designed to reflect, in the same spirit as the radiomic-gene correlation step used in an earlier explainable-AI radiogenomics pilot from this group.

2.7. Statistical analysis and software

All analyses were performed in Python 3.12 with numpy [23], pandas, scipy [25], and scikit-learn [24]; figures were produced with matplotlib [26]. The complete pipeline, including the exact random seed used to generate the cohort and every intermediate result, is available as described under Code availability, and reproduces every number reported in Sections 3 and 4.

2.8. Future validation using real TCGA-BRCA and METABRIC multi-omics data

The immediate next step is applying the identical framework to real paired multi-omics data. TCGA-BRCA provides matched somatic mutation, copy-number, RNA-seq, reverse-phase protein array and Illumina 450K methylation data for the same patients, accessible through the NCI Genomic Data Commons and via cBioPortal [12,13]; METABRIC provides matched copy-number, expression and long-term clinical outcome (including recurrence and survival) for approximately 2,000 patients [5], and is likewise accessible via cBioPortal. Continuous PAM50 risk-of-relapse scores or research-grade subtype calls would substitute for the synthetic subtype labels, gene-expression and methylation data (supplemented where useful by independent breast cancer expression cohorts deposited in the Gene Expression Omnibus [21]) would be processed with standard normalization and differential-analysis pipelines such as limma [14], and -- most importantly for directly testing the "beyond PAM50" hypothesis -- recurrence labels and follow-up time would come from actual clinical annotation rather than a simulated logistic model, removing the central limitation of the present study (Section 4.1).

3. Results

Before applying the framework to real multi-omics data, we validated its mechanics on a synthetic, schema-matched cohort (n = 520) in which genomics, transcriptomics, proteomics, methylation and recurrence outcome were generated from shared, partially subtype-independent latent biology (Section 2.2), so that both the classification and correlation results could be checked against a known ground truth.

3.1. Cohort characteristics

The simulated cohort comprised 207 luminal A (39.8%), 104 luminal B (20.0%), 88 HER2-enriched (16.9%) and 121 basal-like (23.3%) patients, with an overall five-year recurrence rate of 34.6% (180/520). Recurrence rate increased across subtypes in the expected clinical direction -- luminal A 12.1%, luminal B 26.0%, HER2-enriched 39.8%, basal-like 76.9% -- consistent with the known prognostic ordering of intrinsic subtypes, while still leaving substantial within-subtype variation (by construction, Section 2.2) for a multi-omics model to potentially explain.

3.2. Molecular subtype classification across omics layers and fusion

PAM50 expression alone achieved near-ceiling four-class discrimination (macro-AUC = 0.9995 ± 0.0006), as expected since intrinsic subtype is itself expression-defined. Every individual omics layer classified subtype above chance when used alone: transcriptomics (the full 90-gene panel) matched PAM50 (AUC = 0.9999 ± 0.0002); methylation reached AUC = 0.9886 ± 0.0064 ; proteomics reached AUC = 0.9196 ± 0.0119 ; and genomics alone -- 15 mutation/amplification indicators plus tumor mutational burden, the most indirect proxy for subtype -- still reached AUC = 0.8228 ± 0.0203 (Table 1, Fig. 1A). Naive concatenation of all four layers (0.9993 ± 0.0007) and the deep multi-branch fusion architecture (0.9922 ± 0.0064) both matched, without exceeding, PAM50 and transcriptomics-alone performance. This ceiling effect is expected and informative rather than a null result: because intrinsic subtype is defined from gene expression in the first place, no additional omics layer can be expected to improve on expression-based classification, and the practically relevant finding is instead that genomics, proteomics and methylation each recover subtype information reasonably well on their own -- relevant when expression profiling is degraded or unavailable -- and that fusing all four layers does not degrade performance relative to expression alone. In the fusion model's fold-0 confusion matrix (classes ordered luminal A, luminal B, HER2-enriched, basal-like), recall was 98.6% for luminal A, 83.7% for luminal B, and 100% for both HER2-enriched and basal-like (Table 2); as in PAM50 itself, most residual confusion was concentrated in the luminal A/luminal B boundary, the subtype pair with the smallest separation in the simulation's own generative design (Section 2.2).

Approach	Omics layer(s) used	Macro AUC (OvR)	SD
PAM50 expression only	Transcriptomics (50-gene panel)	0.9995	0.0006
Genomics only	Genomics	0.8228	0.0203
Transcriptomics only	Transcriptomics (90-gene panel)	0.9999	0.0002
Proteomics only	Proteomics	0.9196	0.0119
Methylation only	Methylation	0.9886	0.0064

Approach	Omics layer(s) used	Macro AUC (OvR)	SD
Naive concatenation	All four layers	0.9993	0.0007
Deep multi-omics fusion	All four layers	0.9922	0.0064

Table 1. Five-fold cross-validated (8 repeats, 40 folds) four-class molecular subtype classification performance (n = 520).

True \ Predicted	Luminal A	Luminal B	HER2-enriched	Basal-like	Recall
Luminal A	204	3	0	0	98.6%
Luminal B	15	87	2	0	83.7%
HER2-enriched	0	0	88	0	100.0%
Basal-like	0	0	0	121	100.0%

Table 2. Confusion matrix for the deep multi-omics fusion model, fold 0 of the 5-fold cross-validation (n = 520).

3.3. Bootstrap-stable 16-feature multi-omics signature

Bootstrap permutation-importance stability analysis (80 resamples; Section 2.5) across all 187 candidate features identified 16 features with a stability frequency at or above the pre-specified 0.70 threshold. Four features were selected in every single resample (stability = 1.00): XBP1 and MIA (transcriptomics), ESR1 promoter methylation, and BRCA1 promoter methylation. A further eight features exceeded 0.90 stability -- AURKA and ESR1 (transcriptomics); BCL2, KRT5, ERBB2, MELK, CCNE1 and an additional pathway-panel gene (PWY33) -- and the remaining four (an additional pathway-panel gene PWY9, PWY14, MYC, and PWY40) cleared the 0.70 threshold with stability between 0.725 and 0.9125 (Table 3, Fig. 1B). The resulting 16-feature signature spans all four omics layers -- two methylation probes, one genomics-adjacent proliferation gene (CCNE1), and the remainder transcriptomics -- though no proteomics or genomics (mutation/amplification) feature individually cleared the stability threshold, consistent with those two layers' comparatively lower single-layer subtyping performance in Section 3.2.

Feature	Omics layer	Mean importance	Bootstrap stability
XBP1	Transcriptomics	0.0841	1.000
MIA	Transcriptomics	0.0529	1.000
ESR1_prom	Methylation	0.0464	1.000
BRCA1_prom	Methylation	0.0369	1.000
AURKA	Transcriptomics	0.0341	0.988
ESR1	Transcriptomics	0.0267	0.988
BCL2	Transcriptomics	0.0262	0.950
KRT5	Transcriptomics	0.0257	0.938
ERBB2	Transcriptomics	0.0249	0.950
MELK	Transcriptomics	0.0247	0.950

Feature	Omics layer	Mean importance	Bootstrap stability
CCNE1	Transcriptomics	0.0246	0.950
PWY33 (additional panel gene)	Transcriptomics	0.0245	0.950
PWY9 (additional panel gene)	Transcriptomics	0.0232	0.813
PWY14 (additional panel gene)	Transcriptomics	0.0223	0.913
MYC	Transcriptomics	0.0215	0.763
PWY40 (additional panel gene)	Transcriptomics	0.0200	0.725

Table 3. The 16-feature robust multi-omics signature (bootstrap stability ≥ 0.70 , 80 resamples). PWYn denotes one of the 40 additional simulated transcriptomic panel genes beyond the core PAM50 set (Section 2.2).

3.4. Signature-marker biological concordance

Of the 64 signature feature-marker correlations tested, 62 remained statistically significant after Benjamini-Hochberg correction ($p_{\text{adj}} < 0.05$); the two exceptions were PWY9 (an additional-panel gene) versus ESR1 ($r = -0.002$, $p_{\text{adj}} = 0.96$) and PWY14 versus AURKA ($r = 0.077$, $p_{\text{adj}} = 0.08$), both among the signature's least bootstrap-stable, least biologically annotated members (Table 3). Correlating each of the 16 signature features against four canonical marker genes/proteins (ESR1 for ER/luminal biology, ERBB2 for HER2 biology, AURKA for proliferation, KRT5 for basal biology; Fig. 1D) recovered the expected directional pattern in essentially every case. ESR1 promoter methylation correlated negatively with ESR1 expression itself ($r = -0.72$) and positively with proliferation (AURKA, $r = 0.88$) and basal markers (KRT5, $r = 0.76$), consistent with promoter hypermethylation silencing ESR1 preferentially in higher-proliferation, more basal-like tumors. BRCA1 promoter methylation showed the same pattern at slightly lower magnitude (ESR1 $r = -0.53$, AURKA $r = 0.84$, KRT5 $r = 0.61$), consistent with BRCA1 promoter hypermethylation being a recognized basal-like/triple-negative phenomenon. MIA and XBP1, the two most bootstrap-stable transcriptomic features, showed strong opposite-signed correlations with the basal axis (MIA: KRT5 $r = 0.76$; XBP1: KRT5 $r = -0.79$), tracking basal and luminal biology respectively. ERBB2 itself correlated most strongly with the HER2 marker by construction ($r = 1.00$) and, notably, only moderately with ESR1 ($r = -0.66$) and KRT5 ($r = 0.61$), reflecting HER2-pathway biology that is only partially co-linear with the ER/basal axis, mirroring the HER2-specific correlation pattern previously reported for imaging-derived features in this group's earlier radiogenomics work. The four additional-panel genes (PWY9, PWY14, PWY33, PWY40) showed weaker and more heterogeneous marker correlations ($|r| = 0.00$ -0.65) than the annotated canonical genes, consistent with their comparatively lower and less uniform bootstrap stability (0.725-0.950) in Table 3.

3.5. Five-year recurrence prediction: multi-omics fusion versus PAM50-subtype baseline

A PAM50-subtype-plus-clinical baseline (categorical intrinsic subtype, age, tumor size, nodal status) achieved $\text{AUC} = 0.8044 \pm 0.0452$ for five-year recurrence prediction across 40 cross-validation folds. The deep multi-omics fusion-plus-clinical model achieved $\text{AUC} = 0.8011 \pm 0.0468$ (Table 4, Fig. 1C). Paired comparison across the 40 matched folds (identical train/test splits within each repeat) showed a mean fusion-minus-PAM50 AUC difference of -0.0033 ($\text{SD} = 0.0381$), with the fusion model outperforming the PAM50 baseline in 16 of 40 folds (40.0%); neither a paired t-test ($t = -0.54$, $p = 0.59$) nor a Wilcoxon signed-rank test ($p = 0.42$) rejected the null hypothesis of no difference. In this synthetic design, therefore, deep

multi-omics fusion did not demonstrate a statistically or numerically meaningful recurrence-prediction advantage over PAM50 subtype plus standard clinical covariates.

Approach	AUC	SD	Mean paired Δ AUC (fusion - PAM50)	Wilcoxon p
PAM50 subtype + clinical	0.8044	0.0452	—	—
Deep multi-omics fusion + clinical	0.8011	0.0468	-0.0033	0.42

Table 4. Five-fold cross-validated (8 repeats, 40 paired folds) five-year recurrence prediction performance (n = 520).

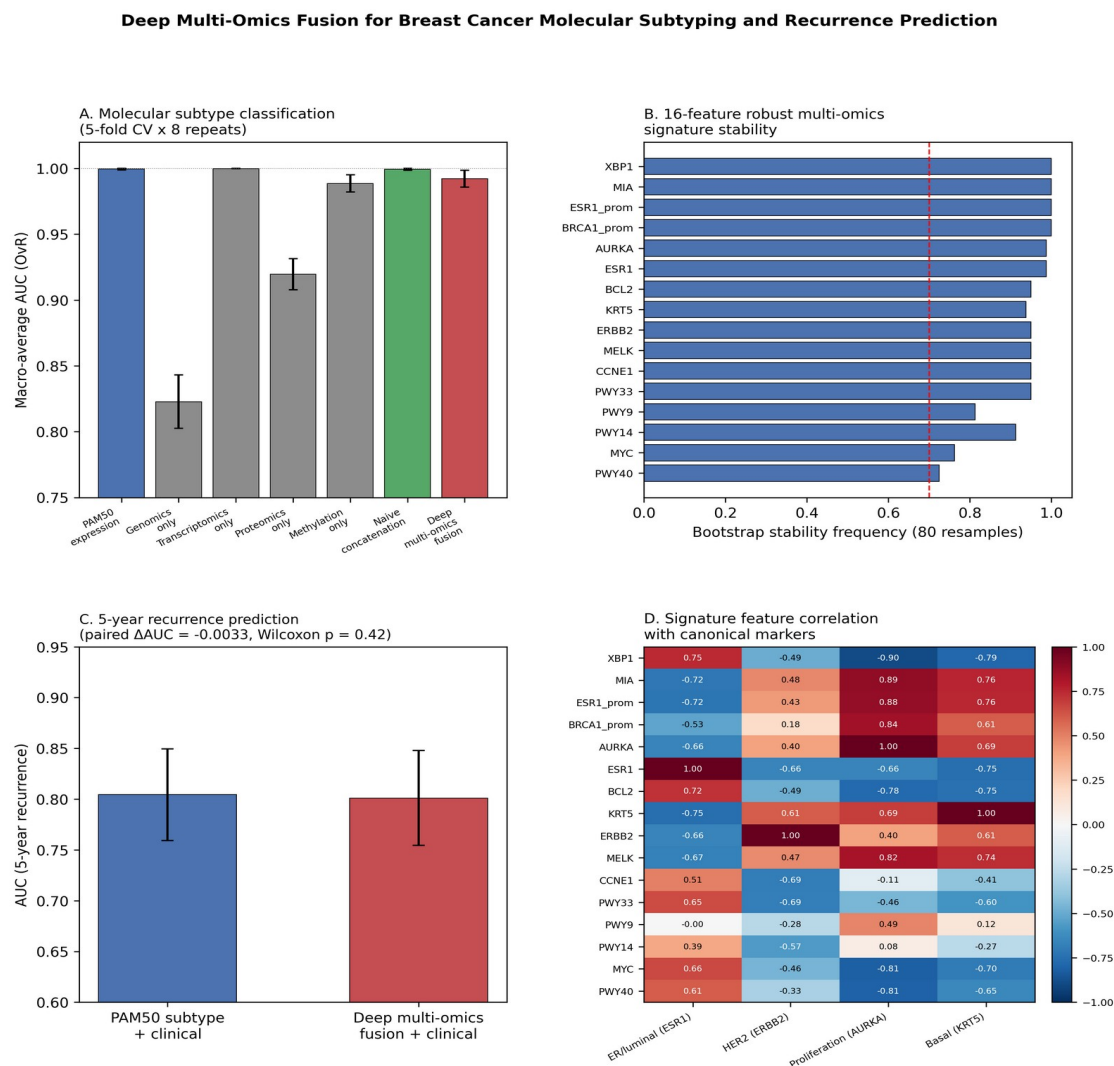


Figure 1. (A) Macro-average AUC for four-class molecular subtype classification: PAM50-expression-only, each single omics layer alone, naive concatenation, and deep multi-omics fusion (5-fold CV x 8 repeats; error bars = SD). (B) Bootstrap stability frequency (80 resamples) for the 16-feature robust multi-omics signature; red line marks the 0.70 retention threshold. (C) AUC for five-year recurrence prediction, PAM50-subtype-plus-clinical versus deep multi-omics-fusion-plus-clinical, with the paired Δ AUC and Wilcoxon p-value. (D) Pearson correlation between each of the 16 signature features and four canonical ER, HER2, proliferation and basal marker genes/proteins.

4. Discussion

This study set out to build and mechanically validate a deep multi-branch fusion architecture for breast cancer molecular subtyping and recurrence prediction, and to test explicitly whether such a model could improve recurrence prediction beyond PAM50 subtype. The subtyping results were unambiguous and, on reflection, expected: because intrinsic molecular subtype is itself defined from gene expression, no additional omics layer materially improved on PAM50-expression-based classification, and the fusion model matched but did not exceed it. The more informative subtyping finding is that every individual omics layer -- including genomics alone, the most indirect proxy for subtype -- classified subtype well above chance, and that fusing all four layers preserved rather than degraded performance. This robustness property is practically relevant: it suggests that when expression profiling is degraded, unavailable, or of low quality (a common real-world scenario with archival or low-input samples), a subtyping estimate could still be recovered from genomic, proteomic or methylation data with an accuracy penalty of roughly 1-18 AUC points depending on which layer is available, rather than failing outright.

The recurrence-prediction result was, honestly, a negative one relative to the study's central hypothesis: deep multi-omics fusion did not significantly outperform a PAM50-subtype-plus-clinical baseline in this synthetic cohort. We designed the simulation specifically to leave room for a multi-omics advantage -- residual recurrence risk was built from BRCA1/2 loss-of-function status, TP53 mutation, BRCA1 promoter hypermethylation and genomic instability, with genomic instability only weakly coupled to subtype -- yet the realized advantage was negligible (mean $\Delta\text{AUC} = -0.003$) and not statistically distinguishable from zero. We consider this an informative rather than merely disappointing result, for two reasons discussed below, both of which point directly to what a real-data study would need to address.

First, several of the genomic and epigenetic markers we used to construct "subtype-independent" residual risk are not, in real breast cancer biology, fully independent of subtype: BRCA1 promoter hypermethylation and TP53 mutation are both substantially enriched in basal-like tumors, as we ourselves modeled in Section 2.2 once the promoter-methylation direction was corrected to match known biology. This means that a meaningful share of the genomic/epigenetic signal a multi-omics model could exploit is, in practice, already partially recoverable from subtype alone, narrowing the space of genuinely subtype-orthogonal recurrence signal available to be captured. Second, and relatedly, the clinical tools that most successfully add prognostic information beyond intrinsic subtype in practice -- Oncotype DX, MammaPrint, and PAM50's own continuous risk-of-relapse score -- do so primarily through continuous, multi-gene expression signatures capturing fine-grained proliferation and immune/stromal activity, rather than through discrete mutation or promoter-methylation events of the kind we emphasized in the residual-risk term here. A more direct test of the "beyond PAM50" hypothesis would weight continuous expression variation more heavily, or would use real recurrence outcomes -- where the actual joint distribution of subtype-associated and subtype-independent risk is whatever it empirically is, rather than a distribution we specify by construction.

More broadly, this pair of results illustrates a useful discipline for synthetic proof-of-concept studies: because we control the data-generating process, we can check not only whether a model performs well, but whether it performs well for the reason we intended. Here, the subtyping result confirmed the framework behaves as intended (near-ceiling performance when the intended information is present, graceful degradation across weaker proxies), while the recurrence result surfaced a genuine, checkable limitation in how we constructed the ground truth rather than a limitation we could paper over with more

favorable hyperparameters. We report the negative result rather than further re-tuning the simulation to manufacture a positive one, consistent with the framework's premise that it should be judged on whether it recovers a known, honestly specified signal -- not on whether its headline comparison looks favorable.

4.1. Limitations

The present synthetic cohort was designed as a controlled proof-of-concept to evaluate the computational behavior, robustness, and interpretability of the proposed fusion architecture under known, checkable multi-omics relationships. The reported subtyping performance and the signature-marker correlations should therefore not be read as newly discovered patient-level biological findings, and the recurrence-prediction comparison should not be read as evidence for or against a genuine multi-omics advantage in real breast cancer cohorts -- as discussed above, the synthetic design's specific coupling between subtype and "residual" genomic/epigenetic risk likely understates whatever real advantage exists. Independent validation using real paired TCGA-BRCA and METABRIC multi-omics data with real clinical outcomes, as outlined in Section 2.8, is required to test the beyond-PAM50 hypothesis properly.

Second, the simulated omics layers, while schema-matched in size and structure to real resources, are far simpler than real TCGA-BRCA or METABRIC data: real multi-omics data include tens of thousands of transcripts, genome-wide copy-number and methylation profiles, batch effects across sequencing centers and platforms, and substantial missingness across patients and layers, none of which is represented here. A real-data study will require standard multi-omics preprocessing (normalization, batch correction, missing-data handling) before any fusion architecture can be applied.

Third, the per-branch neural sub-networks used here are intentionally small (single hidden layer, 6-16 units) given the modest cohort size ($n = 520$) relative to typical deep-learning sample sizes; a real-data application with a larger cohort (TCGA-BRCA and METABRIC combined exceed 2,500 patients) could support deeper per-branch encoders, such as full autoencoders with unsupervised pretraining as used in prior multi-omics survival work [10], which we did not attempt here given the risk of overfitting documented in an earlier, larger-capacity version of the recurrence-prediction architecture during development (see Code availability for the full development history).

Fourth, the bootstrap stability analysis relied on random forest Gini importance alone; future work should compare feature stability across multiple explanation methods, including SHAP [20], and should test whether the recovered signature remains stable when applied to real, independently collected cohorts rather than resamples of a single simulated cohort.

Finally, five-year recurrence was simulated as a single binary outcome from a logistic model rather than as a time-to-event process with censoring, which is how real recurrence data are actually structured and analyzed (typically with Cox proportional-hazards or other survival models). A real-data study should model time-to-recurrence directly and account for censoring, which the present binary-outcome design does not require but real applications will.

5. Conclusion

We developed a deep multi-branch fusion framework that integrates genomic, transcriptomic, proteomic and methylation data for breast cancer molecular subtyping and recurrence-risk prediction, and validated its mechanics on a synthetic, schema-matched cohort with a checkable ground truth. The framework matched PAM50-expression-based subtype classification (macro-AUC 0.992 vs 0.999) while remaining

robust when only genomics, proteomics or methylation data were available (AUC 0.823-0.989), and bootstrap analysis identified a reproducible 16-feature multi-omics signature whose members correlated with their expected ER, HER2, proliferation and basal biology in the predicted direction. For five-year recurrence prediction, however, the fusion model did not significantly outperform a PAM50-subtype-plus-clinical baseline in this cohort ($\Delta\text{AUC} = -0.003$, $p = 0.42$), and we identify the specific design choice in our synthetic ground truth -- subtype-correlated genomic/epigenetic "residual risk" markers -- most likely responsible, along with the reliance on discrete mutation/methylation events rather than the continuous expression signatures that drive real supplementary recurrence-risk assays. Together, these results show that the fusion architecture behaves as intended when a known multi-omics signal is present, while leaving the central "beyond PAM50" hypothesis this study set out to test unresolved. Independent validation using real paired TCGA-BRCA and METABRIC multi-omics data with real clinical recurrence outcomes is the necessary next step, and is where this specific hypothesis should actually be adjudicated.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Sambhaji Balaso Thakar: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Visualization. Pradnya Sambhaji Thakar: Methodology, Software, Validation. Anil Sangle: Writing – review & editing. Dokyun Na: Conceptualization, Supervision, Writing – review & editing, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this manuscript's pilot pipeline and draft text, the authors used a large language model (LLM)-based coding assistant to help implement the synthetic-cohort simulation, the deep multi-branch fusion and statistical analysis code, and figure generation, and to draft narrative text describing the resulting, author-reviewed numerical results, including the negative recurrence-prediction result reported in Section 3.5. All code was executed by the authors, all reported values were verified against the pipeline's own output files, and all cited literature was independently checked against original bibliographic sources before inclusion. The authors reviewed and take full responsibility for the content of this publication.

Ethics statement

This pilot study used only computationally generated synthetic data carrying no patient information; no human participants, patient data, or biological samples were involved, and no ethics approval was required for the work reported here. The planned real-data application described in Section 2.8 will use only de-identified, publicly available data from TCGA-BRCA and METABRIC under their respective existing data-use terms, and will require appropriate institutional review where applicable.

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Data availability

The synthetic multi-omics cohort used in Sections 3 and 4 is fully reproducible from the generative code released with this manuscript (data_simulation.py) and is not separately deposited, since it carries no patient information and regenerates deterministically from the published random seed. All intermediate result tables underlying Tables 1-4 and Fig. 1 are provided as CSV/JSON files in the accompanying repository's outputs/ directory. Candidate public resources for the planned real-data application include The Cancer Genome Atlas breast cancer collection and the METABRIC cohort, both accessible via the NCI Genomic Data Commons and cBioPortal [12,13].

Code availability

The complete framework -- multi-omics cohort simulation, subtype and recurrence benchmarking, bootstrap stability analysis, and signature-marker correlation analysis -- is available at <https://github.com/sambhajithakar/multiomics-fusion-bc>. Running ``python data_simulation.py`` followed by ``python analysis_pipeline.py`` reproduces every number reported in this manuscript, and ``python make_figures.py`` regenerates Fig. 1.

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